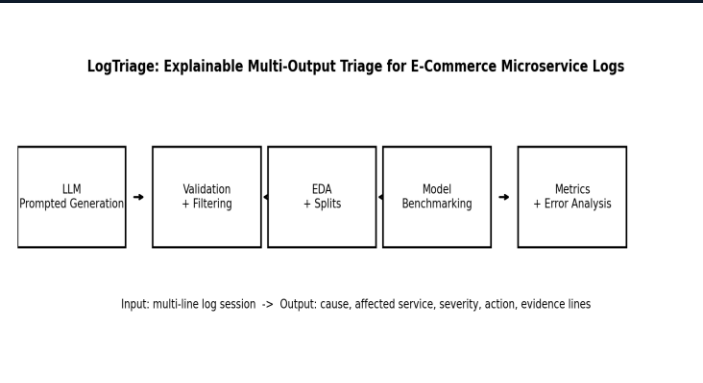


Project review

Project motivation

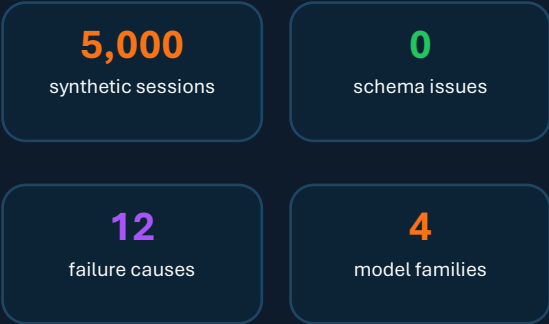
E-commerce checkout failures produce logs across multiple services.
Operators need structured triage: cause, service, severity, action, and evidence.
The task is explainable multi-output NLP, not just anomaly detection.



What changed from proposal

Added affected_service as a prediction output.
Expanded the fixed taxonomy to 12 causes and 12 actions.
Created 5,000 examples instead of a small prototype.
Added validation, EDA visuals, and baseline evaluation.

Current contributions



Novelty

The benchmark predicts operator-ready triage with evidence lines, not only normal/abnormal labels.

Previous work

Related work table

Paper / year	Task	Methods	Data	Relation to LogTriage
LogLLM (2024)	Log anomaly detection	LLM-based log understanding	Public system logs	Shows LLMs can capture log semantics; our task adds triage + evidence.
AIOps benchmark datasets (2022)	Anomaly/RCA benchmarks	Large-scale benchmark construction	Real-world AIOps data	Motivates benchmark design; our dataset is synthetic and explainable.
Synthetic data generation survey (2024)	LLM synthetic data quality	Generation, curation, filtering	Text benchmarks	Supports controlled generation + validation methodology.

Takeaway: prior work covers anomaly detection and RCA, so LogTriage is framed narrowly as explainable multi-output operator triage with action and evidence-line prediction.

Dataset

Generated dataset

5,000

examples

3,500

train

750

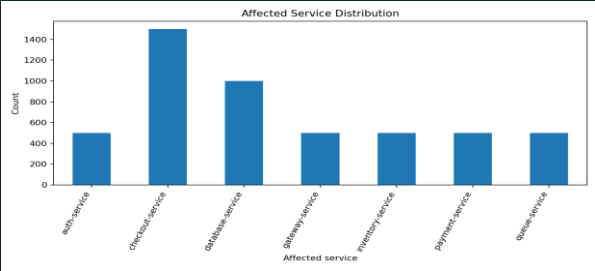
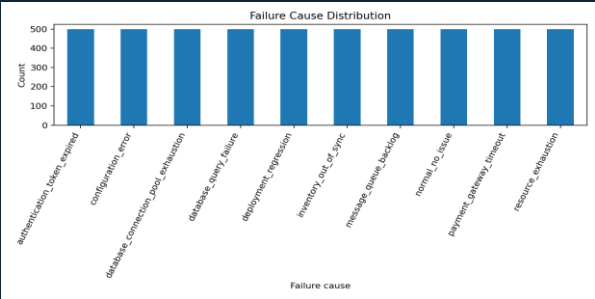
valid

750

test

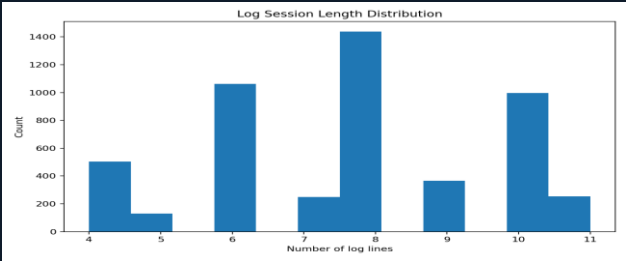
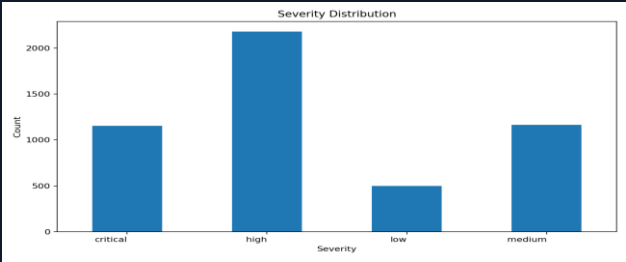
Each record includes numbered logs and structured labels. Labels are generated using controlled attributes and fixed taxonomies. Validation checks JSON schema, allowed labels, evidence-line numbers, and duplicates.

EDA: label coverage



Cause and service distributions are checked to avoid one-label dominance.

EDA: input quality



Severity, noise, and session length are tracked because they affect model difficulty.

Baseline solution and results

Baseline methods

Rule-based baseline: transparent keyword/runbook rules.
TF-IDF + Logistic Regression / Linear SVM: classical NLP baseline.
LLM prompt baselines prepared as zero-shot/few-shot evaluation scripts.
Transformer training script prepared for final extension.

Baseline goal

Measure how much can be solved before expensive fine-tuning.

Results table

Model	Cause	Sev.	Action
Rule-based	0.920	0.905	0.897
TF-IDF + LogReg	0.961	0.952	0.959
Zero-shot LLM	prepared	prepared	prepared
Few-shot LLM	prepared	prepared	prepared

Current completed results focus on implemented baselines. LLM/transformer runs are prepared for final evaluation or future extension.

Visual + errors



Inspected error types

timeout vs external API confusion
symptom lines selected instead of root cause
severity depends on context and impact
synthetic wording may create lexical shortcuts

Plan

Remaining project plan

Step	Scope	Technical work	Expected outcome	Status
1	Data hardening	Add harder noise and misleading symptom lines	More realistic test set	next
2	LLM baselines	Run zero-shot/few-shot API evaluation	LLM comparison table	next
3	Transformer	Fine-tune DistilBERT/RoBERTa classifier	Fine-tuned benchmark	next
4	Error analysis	Inspect wrong examples by class and noise level	Explain results	in progress
5	Final presentation	Prepare final 5-slide deck + report artifacts	Submit/present final project	last

Expected final outcome: a complete course package with task definition, synthetic data recipe, validation, EDA, model benchmarking, evaluation, error analysis, slides, code, data, and report.